UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN
Department of Electrical and Computer Engineering

ECE 310 Digital Signal Processing – Fall 2026

Homework 5

Profs. Kamalabadi, Snyder **Due: Sunday, Oct. 4, 2025, 11:59pm on Gradescope**

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1. Compute the discrete-time Fourier transform (DTFT) for each of the following discrete-time sequences. **Do not use DTFT table pairs unless noted otherwise.** You may use DTFT table properties, if necessary.

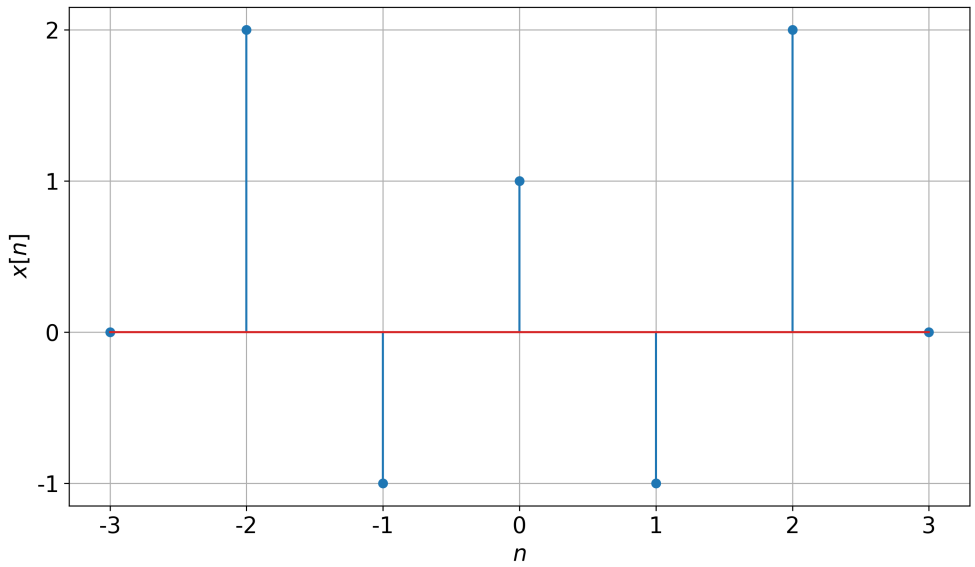
   (a) $x[n] = \{1, 0, \underset{\uparrow}{0}, 0, -1\}$

   (b) $x[n] = u[n] - u[n-4]$

   (c) $x[n] = \cos\left(\frac{\pi}{3}n + \frac{\pi}{4}\right)$, you may use $e^{j\omega_0 n} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega - \omega_0)$, if necessary.

   (d) $x[n] = \alpha^n e^{j\omega_0 n} u[n]$, where $|\alpha| < 1$ and ω_0 is a real constant. You may use $e^{j\omega_0 n} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega - \omega_0)$, if necessary.

2. A discrete-time signal $x[n]$ is depicted in the below figure. Without explicitly determining a closed-form expression for $X_d(\omega)$, find each of the following quantities.



   (a) $X_d(0)$

   (b) $X_d(\pi)$

   (c) $\int_{-\pi}^{\pi} X_d(\omega) d\omega$

   (d) $\int_{-\pi}^{\pi} |X_d(\omega)|^2 d\omega$